

PRD - Units Page

Combined inventory, sales visibility, bulk management and project-wide unit operations

Prepared for Full Stack Product, Design, Backend, Frontend, Data, QA and DevOps teams

Document	PRD - Units Page
Version	1.0
Date	2026-08-04
Status	Implementation-ready functional PRD
Access model	Phase 1: every authenticated user can view and use every feature

Role restrictions and field-level visibility will be introduced in a later permissions PRD. The implementation must remain permission-ready while defaulting to allow-all behavior in this phase.

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1. Executive Summary

The Units Page is the central inventory workspace for every sellable, rentable, assignable or trackable unit inside a project. It combines apartments, offices, retail spaces, parking spaces, storage rooms, villas, hotel rooms, land plots and custom unit types in one page. The page is optimized for rapid scanning, advanced filtering, comparison, sales operations, finance visibility and bulk management.

Confirmed product direction: type summary cards at the top, followed by one combined units view. Table view is the default; card view is optional. There are no separate unit-type tabs.

A project may use either Project -> Units or Project -> Structure/Building -> optional Floor -> Units. The same page must support both hierarchies without forcing an empty structure or floor.

2. Product Goals and Non-Goals

Goals

- Give users one complete source of truth for all units in a selected project.
- Make availability, pricing, buyer and payment state understandable without opening each Unit Page.
- Support projects ranging from a few units to tens of thousands of units.
- Allow safe bulk creation, editing, import, export, status changes and structure assignment.
- Support simple and complex project structures with nullable structure and floor relationships.
- Remain generic enough for apartments, offices, parking, retail, storage, villas, hotel rooms, land plots and custom unit types.
- Provide a stable platform contract for Sales, Finance, Clients, Documents, 3D, Contracts and Project modules.

Non-Goals for this PRD

- Detailed design of the Unit Page, which is defined in a separate PRD.
- Role-based visibility and edit restrictions. Every authenticated user has full access in this phase.
- Detailed accounting ledger behavior. Financial values are displayed from linked Finance records.
- Detailed contract authoring and e-signature workflows.
- Final visual design tokens. The page must follow the existing premium platform design system.

3. Access Model for This Phase

Phase 1 access rule: every authenticated platform user can view, create, edit, import, export, hold, reserve, sell, release, split, merge and manage all unit information. No feature or field is hidden based on role in this PRD.

The implementation must still be permission-ready. Frontend actions and backend endpoints should call a common authorization interface, but the default policy for this phase returns ALLOW for every authenticated user. A later Permissions PRD will replace the default policy without requiring a page rewrite.

Capability	Current behavior	Future-ready implementation
View commercial and client data	Allowed for all authenticated users	Wrap with capability key units.view_commercial
Create/edit/delete/archive	Allowed for all authenticated users	Wrap with units.manage
Bulk import/export	Allowed for all authenticated users	Wrap with units.bulk_manage
View and edit finance summaries	Allowed for all authenticated users	Wrap with units.view_finance / units.edit_finance
Hold/reserve/sell/release	Allowed for all authenticated users	Wrap with units.sales_actions

4. Information Architecture

Supported hierarchy:

- Simple project: Project -> Units -> Unit Page.
- Structured project: Project -> Building / Block / Tower / Phase / Zone -> optional Floor / Level -> Units -> Unit Page.

The Units Page is opened from the Project Page and always receives a project context. It may optionally receive a structure or floor context to pre-filter results.

Entry point	Behavior
Project Page -> Units	Open all units in the project.
Building/Structure Page -> Units	Open Units Page pre-filtered to the selected structure.
Floor/Level Page -> Units	Open Units Page pre-filtered to the selected floor.
Global Search result	Open Unit Page directly or open Units Page with a matching search filter.
3D Project Viewer	Open the selected Unit Page or Units Page with selected 3D context.

5. Page Layout and Responsive Behavior

Desktop layout

1. Page header with breadcrumb, project identity, unit count and top-right actions.
2. Type summary card strip.
3. View toolbar with table/card toggle, filters, sort, column selection, export and bulk action entry points.
4. Combined units table or card grid.
5. Selection action bar when one or more units are selected.
6. Pagination or virtualized infinite loading area.

Mobile layout

- Summary cards become a horizontal swipe row.
- Default view becomes compact cards or a horizontally scrollable table according to the selected preference.
- Filters open in a full-height bottom sheet.
- Bulk selection remains available with a sticky action bar.
- Critical fields shown first: unit code, type, status, floor, area, total price, buyer/payment state.

UNITS-001 - The page must preserve the selected view, visible columns, sort and filter state per user and project.

UNITS-002 - The page must remain usable at 320 px viewport width and must not rely on hover-only actions.

6. Unit Type Summary Cards

The top section shows one summary card for each configured unit type. Standard types include Apartments, Offices, Retail, Parking, Storage, Villas, Hotel Rooms and Land Plots. Administrators can add custom unit types later; the Units Page must render them dynamically.

Card element	Requirement
Type name and icon	Use configured display name and icon.
Total count	Count all active units of the type.
Available	Count units with Available status.
Reserved	Count On Hold and Reserved separately in the API; UI may show Reserved total with detail tooltip.
Sold/Assigned	Count Contracted, Sold, Handed Over, Company-Owned or Rented according to configured summary mapping.
Interaction	Clicking a card applies or removes the corresponding unit-type filter.

UNITS-003 - Summary counts must respect the current project, structure, floor and all active filters except the unit-type filter represented by the card itself.

UNITS-004 - Counts must update after create, import, status transition, split, merge or delete/archive without requiring a full page reload.

7. Combined Units Table and Card View

Default table columns

Column	Source / behavior
Selection	Checkbox; select page, visible result set or explicit filtered result set.
Unit code	Unique inside project or configured structure scope; opens Unit Page.
Type	Configured unit type.
Structure	Building/block/tower/phase/zone; blank for direct-to-units projects.
Floor/Level	Optional; numeric or configured label.
Internal area	Decimal m2.
Balcony/Terrace area	Decimal m2; blank/zero if absent.
Total area	Sum of physical area components.
Rooms/Bedrooms	Dynamic according to unit type.
Status	Lifecycle status badge.
Main price/m2	Current internal/main area price per m2.
Final total price	Calculated current final price.
Client/Buyer	Linked active buyer or reservation client.
Amount paid	Read model from Finance/Payments.
Remaining amount	Final sale value minus posted payments.
Next payment	Amount and due date.
Overdue	Warning badge when any scheduled payment is overdue.
Actions	Open, edit, duplicate, hold/reserve, change status, archive.

Card view

- Unit image or floor-plan thumbnail.
- Unit code, type, structure/floor and status.
- Internal and balcony area.
- Current final price and main price/m2.
- Buyer/payment summary where applicable.
- Quick actions identical to table view.

UNITS-005 - Changing between table and card view must not reset filters, selection or pagination cursor.

UNITS-006 - Users can choose visible columns and reorder supported columns. Required identity columns cannot be removed simultaneously.

8. Search, Filters, Sorting and Saved View State

The platform Global Search remains the universal search entry. Inside the Units Page, the page consumes global query state and exposes an advanced Filters control suited to units. There is no second independent search index.

Filter group	Supported values
Type	One or more unit types.
Lifecycle status	Draft, Available, On Hold, Reserved, Contracted, Sold, Handed Over, Not for Sale, Company-Owned, Rented.
Location hierarchy	Project, structure/building/block/tower/phase/zone, floor/level.
Area	Internal area, balcony area, total area and configurable component ranges.
Commercial	Main price/m2, final price, buyer, sales agent, reservation expiry.
Payments	Fully paid, partially paid, unpaid, overdue, next payment date range.
Type-specific	Bedrooms, bathrooms, orientation, parking category, retail frontage, land zoning and dynamic custom fields.
Record state	Active, archived, incomplete, validation warnings.

Default sort is pinned operational priority: On Hold expiring soon, overdue payments, Available, Reserved, Contracted, Sold/Handed Over, then unit code. Users may replace the sort.

UNITS-007 - Filter state must be serialized into the URL so links can be shared and browser navigation works correctly.

UNITS-008 - The backend must perform filtering and sorting; the frontend must not fetch the complete project inventory for client-side filtering.

9. Bulk Actions

Bulk actions are available after selecting units. In this phase, all authenticated users can run all actions.

Bulk action	Behavior and safeguards
Edit common fields	Update type-compatible common fields; show preview and affected count.
Assign structure/floor	Assign or clear optional structure and floor. Validate project ownership.
Change status	Validate every unit against allowed transitions; show per-unit failures.
Update main price/m2	Recalculate final price using each unit area and component pricing.
Update component price/m2	Update selected area component such as balcony for eligible units.
Add/remove custom tag	Apply operational tags.
Archive	Soft archive with confirmation and dependency warning.
Export selected	Export current selected records and visible/selected fields.
Create linked package	Add selected units to a sales package.

UNITS-009 - Bulk actions must use an asynchronous job when more than 100 records are affected.

UNITS-010 - Bulk jobs must return success, warning and failure counts with an item-level downloadable report.

UNITS-011 - Bulk operations must be idempotent when retried using the same idempotency key.

10. Create, Duplicate and Import Workflows

Create New Unit

- Top-right Create New Unit action.
- Opens Create New Unit flow with project context preselected.
- Structure and floor remain optional.
- Unit type determines dynamic field set.
- Area components and pricing can be configured during creation.
- Save as Draft or create directly in Available/Not for Sale state.

Duplicate Unit / Unit Template

- Duplicate one unit into one or many new units.
- Allow code pattern generation, floor increment and structure assignment.
- Copy physical fields, area components, type-specific fields, documents/visual references optionally.
- Never copy buyer, reservation, payment, contract or handover state.

Bulk Import Units

7. Download project-specific import template with configured unit types and custom fields.
8. Upload CSV/XLSX file.
9. Map columns when headers do not match.
10. Run dry-run validation without writing records.
11. Show row-level errors and warnings.
12. Allow corrected file upload or commit valid rows only.
13. Commit creates an import batch and processes asynchronously.
14. Display progress and downloadable result report.

Import validation	Rule
Unit code	Required and unique inside configured project/structure scope.
Unit type	Must match configured type or explicit custom type creation option.
Project hierarchy	Structure/floor must belong to the same project; floor must belong to structure when both exist.
Areas	Non-negative decimal values; internal area required for applicable types.
Pricing	Decimal currency values; total must not be lower than non-main component subtotals.
Status	Must be recognized and valid for new record.

Import validation	Rule
Dynamic fields	Required type-specific fields enforced.
Duplicate rows	Duplicates inside file detected before commit.

UNITS-012 - Import commit must be atomic per row, not all-or-nothing for the complete file. The result must clearly identify rejected rows.

UNITS-013 - The same file committed twice with the same import batch id must not create duplicates.

11. Export and Reporting

- Export current filtered result set, current page or selected units.
- Formats: XLSX and CSV; PDF summary may be added later.
- Include visible columns by default with option to include all standard and dynamic fields.
- Financial values use current currency and decimal precision.
- Export metadata includes project, filters, generated by, generated at and record count.
- Large exports run asynchronously and appear in the user notification/download center.

UNITS-014 - Exports must use the same backend query contract as the page to prevent differences between displayed and exported results.

12. Linked Unit Packages, Split and Merge

Linked Unit Package

A commercial package can contain an apartment, one or more parking spaces, storage, garden rights, furniture or finish package, while every physical unit remains a separate record.

Rule	Requirement
Package identity	Unique package id, name, project, currency and status.
Items	Unit records and non-unit add-ons with quantity and price allocation.
Availability	Package cannot reserve/sell the same physical unit in two active packages.
Pricing	Package total may equal item sum or include explicit package adjustment.
Navigation	Units Page shows package badge and opens Linked Unit Package Setup.

Unit Split

- Create two or more successor units from one source unit.
- Source is locked during operation.
- Areas and price allocation must reconcile to source totals unless user confirms a valid adjustment.
- Active reservation, contract or posted payment blocks split until resolved.
- Source becomes Split/Archived and retains links to successors.

Unit Merge

- Create one successor unit from two or more compatible units in the same project.
- All source units must be free from conflicting active reservations/contracts/payments.
- Merged areas and price components are reviewed before commit.
- Source units become Merged/Archived and retain link to successor.

13. Status, Financial and Payment Visibility

The Units Page displays operational read models. The Unit domain owns physical identity and lifecycle status; Sales owns reservation/contract context; Finance owns posted payments and schedules. The page composes these values without creating duplicate financial truth.

Displayed value	Authoritative source
Availability/lifecycle status	Unit domain and active hold/reservation state.
Buyer/client	Active reservation/contract relationship.
Final sale value	Active contract or current unit price when unsold.
Amount paid	Posted Finance transactions allocated to unit/contract.
Remaining amount	Contract value less posted payments and approved credits.
Next payment	Payment schedule.
Overdue warning	Payment schedule due date compared with current date and posted amount.

UNITS-015 - Financial summaries must be read-only projections on the Units Page unless the user opens the relevant payment or contract workflow.

14. Validation, Concurrency and Error Handling

Scenario	Required behavior
Duplicate unit code	Reject write with field-level error and conflicting unit link.
Concurrent edit	Use optimistic concurrency version; show changed fields and reload/overwrite choice.
Concurrent hold/reservation	Database constraint/transaction permits only one active exclusive hold/reservation per unit.
Partial bulk failure	Commit valid units; report failures with reasons.
Finance service unavailable	Show unit inventory with payment fields marked temporarily unavailable; do not block page.
3D mapping unavailable	Hide/disable 3D locator only; page remains functional.
Archived dependency	Prevent destructive delete; use soft archive and dependency details.
Invalid total price	Reject if entered total is below the subtotal of fixed/non-main priced components.

Price edits update the current pricing snapshot only. The product does not maintain a dedicated asking-price history or show price changes in the Unit Activity Timeline.

15. Functional Data Model

Entity	Key fields and constraints
Unit	id UUID; tenant/company/project ids; nullable structure_id and floor_id; unit_code; type_id; lifecycle_status; currency; current totals; version; timestamps; archived_at.
UnitType	id; project/global scope; code; display name; icon; active flag; schema version.
UnitFieldDefinition	type_id; field key; label; data type; required flag; validation; sort/filter metadata.
UnitFieldValue	unit_id; definition_id; typed value or JSON value; unique unit+definition.
UnitAreaComponent	unit_id; component type; label; area_m2; price_per_m2; subtotal; sort order; included_in_total.
UnitReadModel	Denormalized query projection containing hierarchy, status, client and payment summary for fast page reads.
UnitImportBatch	file metadata; mapping; status; counts; idempotency key; result report.
UnitPackage	project; name; status; currency; total; adjustment.
UnitPackageItem	package; unit/add-on ref; quantity; allocated price; active exclusivity metadata.
UnitStructureOperation	split/merge type; source ids; successor ids; reconciliation data; status.
UnitSavedView	user; project; filters; sort; visible columns; view mode.

Money and numeric rules

- Use fixed precision Decimal/Numeric database types; never Float for money or area calculations.
- Store currency ISO code on unit and financial record.

- Round component subtotal and final total using configured currency precision and deterministic rounding mode.
- Store areas as Decimal with project-configured scale, recommended 4 decimal places.

16. API Contract

Method and endpoint	Purpose
GET /projects/{projectId}/units	Paginated/virtualized units query with filters, sort, columns and cursor.
GET /projects/{projectId}/units/summary	Unit-type and lifecycle summary counts.
POST /projects/{projectId}/units	Create one unit.
POST /units/{unitId}/duplicate	Duplicate one unit or generate many from a template.
PATCH /units/{unitId}	Edit current unit data with version precondition.
POST /units/bulk-update	Start bulk update job.
POST /unit-imports/dry-run	Validate and map import without writes.
POST /unit-imports/{batchId}/commit	Commit validated import.
GET /unit-imports/{batchId}	Progress and result.
POST /unit-exports	Start export job.
POST /unit-packages	Create linked package.
POST /unit-structure-operations/split	Start controlled unit split.
POST /unit-structure-operations/merge	Start controlled unit merge.
POST /units/{unitId}/archive	Soft archive.

Query response should return a stable row schema with standard fields, dynamic_fields, pricing_summary, sales_summary and payment_summary. Endpoints must accept idempotency keys for creates and asynchronous jobs.

17. Frontend Architecture

- Route: /projects/:projectId/units with optional structureId, floorId and serialized filter query parameters.
- Use server-driven pagination/cursor and virtualization for large datasets.
- Keep page query state in URL plus a per-user saved-view preference.
- Use normalized client cache keyed by unit id and query key.
- Subscribe to unit/status/import/bulk-job events through WebSocket or SSE to refresh affected rows and summary cards.
- Use a reusable dynamic-field renderer based on UnitType schema.
- Keep selection state explicit; distinguish current-page selection from all-filtered-result selection.
- All destructive or high-impact actions require review/confirmation with affected count.

Frontend component	Responsibility
UnitsPageShell	Route context, header, data orchestration and error boundary.
UnitTypeSummaryCards	Counts and type filter behavior.
UnitsToolbar	View toggle, filters, sort, columns, export, create/import.
UnitsTable	Virtualized rows, selection, inline badges and navigation.
UnitsCardGrid	Responsive visual alternative.
UnitFiltersDrawer	Dynamic standard and type-specific filters.
BulkActionBar	Selection-aware actions and job progress.
ImportWizard	Upload, map, dry-run, commit and report.
SavedViewController	Persist per-user view state.

18. Backend and Integration Architecture

- Unit Service owns unit identity, hierarchy, type-specific data, area components and lifecycle state.
- Unit Query Service/read model composes Unit, Sales, Client and Finance summaries for fast list access.
- Sales integration provides holds, reservations, buyer and contract context.
- Finance integration provides posted paid amount, remaining amount, schedule and overdue state.
- Document integration provides thumbnails and linked document counts.
- 3D integration maps unit id to model object/feature id.
- Background Worker executes imports, exports and large bulk operations.

- Object storage stores import/export files and reports.

Domain events

Event	Consumers
unit.created / unit.updated / unit.archived	Read model, search index, Project summary, 3D sync.
unit.status_changed	Sales, Project dashboard, notifications, read model.
unit.pricing_updated	Read model only; no business price-history timeline.
unit.import_completed	Notifications and Units Page refresh.
unit.bulk_job_completed	Notifications and result center.
unit.package_changed	Sales and contract package context.
unit.split_completed / unit.merge_completed	Documents, 3D, Sales and read model.
payment.posted / payment.schedule_changed	Unit read model.

19. Non-Functional Requirements

Area	Target
Performance	First usable page under 2.5 s at p75 on normal broadband; filter response under 800 ms p75 for indexed queries.
Scale	At least 50,000 units per project and 1,000,000 units per tenant without client-side full load.
Availability	Page read path target 99.9% monthly; degraded finance/3D integrations must not block inventory access.
Accessibility	WCAG 2.1 AA target; full keyboard support; labels and status not color-only.
Localization	Locale-aware numbers, dates, currencies, area units and translated unit-type labels.
Security baseline	Authenticated access, input validation, CSRF/XSS protections, encrypted transport, object-storage signed URLs.
Data integrity	Database uniqueness, foreign keys, optimistic concurrency and idempotency.
Browser support	Current and previous major Chrome, Edge, Safari and Firefox; responsive mobile browsers.

20. Analytics and Observability

- Page viewed, filter applied, summary card selected, view toggled, column changed.
- Create, duplicate, import dry-run, import commit, export, bulk action start/complete.
- Hold/reserve/sell/release entry from list.
- API latency, query cardinality, cache hit, job duration and failure rate.
- Structured logs include tenant, company, project, request id, user id and job id, but no sensitive client values in plain text.
- Alert on import failure spikes, duplicate conflicts, query timeout and read-model lag.

21. QA Strategy and Test Matrix

Test area	Minimum coverage
Unit tests	Pricing helpers, filter serialization, validators, status mapping, dynamic-field rendering.
API integration	CRUD, list filters, cursor pagination, idempotency, optimistic concurrency, import/bulk jobs.
Database	Uniqueness, hierarchy constraints, decimal precision, split/merge reconciliation.
End-to-end	Create, filter, edit, import, export, bulk update, hold/reserve navigation, mobile use.
Concurrency	Two users edit same unit; two users attempt hold/reservation; repeated import commit.

Test area	Minimum coverage
Performance	50k-unit project list, complex filters, large export/import.
Accessibility	Keyboard, screen reader labels, focus management, contrast and status semantics.
Failure injection	Finance unavailable, worker retry, object storage failure, partial bulk error.

22. Acceptance Criteria

- AC-U-01** - The page opens from a project and displays all accessible project units in one combined view.
- AC-U-02** - Summary cards show unit-type totals and filter the combined result.
- AC-U-03** - Table is default and card view is optional without losing state.
- AC-U-04** - Simple projects work without structure/floor; complex projects can filter by both.
- AC-U-05** - Current price, client, paid, remaining and next-payment information are visible for all users in this phase.
- AC-U-06** - Users can create, duplicate, import, export and bulk edit units.
- AC-U-07** - Import supports mapping, dry-run, row-level validation and asynchronous commit.
- AC-U-08** - Bulk actions return item-level success and failure results.
- AC-U-09** - Duplicate unit codes are prevented by database constraint and user-facing validation.
- AC-U-10** - Hold/reservation exclusivity is concurrency-safe.
- AC-U-11** - Component-based pricing values and final price are shown consistently.
- AC-U-12** - Linked packages, split and merge flows are reachable from the page.
- AC-U-13** - The page is responsive, keyboard accessible and works with large inventories.
- AC-U-14** - All features are enabled for every authenticated user while permission hooks remain in place.

23. Page and Flow Registry

Name	Purpose / status
Units Page	Defined by this PRD.
Unit Page / Unit Profile	Defined by separate PRD.
Create New Unit	Supporting creation flow.
Bulk Import Units	Upload, map, validate and commit units.
Bulk Unit Edit	Selection-based common updates.
Unit Template / Duplicate Unit	Generate similar units safely.
Unit Hold Flow	Temporary exclusive hold.
Unit Reservation Flow	Reservation conversion and management.
Unit Status Change Flow	Validated lifecycle transition.
Linked Unit Package Setup	Package apartment/parking/storage/add-ons.
Unit Split Flow	Controlled source-to-successors operation.
Unit Merge Flow	Controlled sources-to-successor operation.
Unit Area & Pricing Setup	Configure area components and current pricing.
Area Component Editor	Edit component area and price/m2.
Pricing Calculation Breakdown	Show component subtotals and final total.
Unit Handover	Inspection and handover workspace.
Unit Activity Timeline	Operational history excluding asking-price history.